

Algebra II | Guided notes — we fill these in together

Vocabulary

Maximum Value: The greatest output of a function; occurs at the vertex when a is negative.

Minimum Value: The least output of a function; occurs at the vertex when a is positive.

Optimization: Using a quadratic model to find the best possible (largest or smallest) outcome of a real situation.

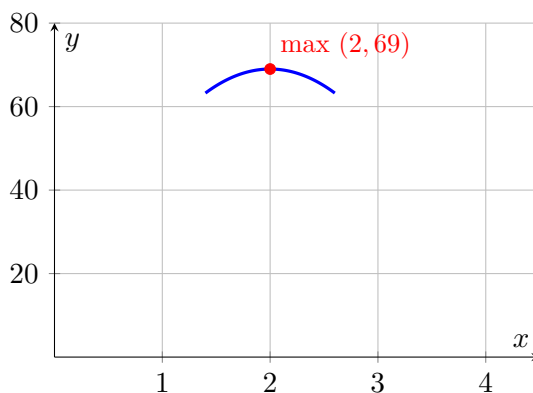
We do together. A ball's height is modeled by $h(t) = -16t^2 + 64t + 5$. Find the maximum height and when it occurs.

Step 1. Since $a = -16$ is negative, the function has a maximum.

Step 2. Find the time: $t = -\frac{b}{2a} = -\frac{64}{2(-16)} = \underline{2}$.

Step 3. Substitute $t = 2$: $h(2) = -16(2)^2 + 64(2) + 5 = -64 + 128 + 5 = \underline{69}$.

Step 4. The maximum height is 69 feet, occurring at $t = \underline{2}$ seconds.



Height model $h(t) = -16t^2 + 64t + 5$ reaching a maximum of 69 ft at $t = 2$ s.

You Try. Solve each. Show every step, then name the answer.

- a) A company's profit is modeled by $P(x) = -3x^2 + 24x - 20$, where x is the number of items (in hundreds) sold. Find the maximum profit and the number of items that produce it. $x = -\frac{24}{2(-3)} = 4$;
 $P(4) = -3(16) + 96 - 20 = 28$. Maximum profit is 28 (hundred dollars) when 400 items are sold.

Summary (fill in): The maximum or minimum value of a quadratic function is the y-coordinate of the vertex, k , which occurs at $x = \underline{h}$. If a is negative the function has a maximum; if a is positive it has a minimum. In real-world problems, always interpret the answer with correct units.